

LangChain, LangGraph & FastAPI Mastery Plan

A 6-Week Comprehensive Learning Path for Production AI Systems

Learning Roadmap Overview

Phase 1: Foundation (Week 1-2)

- LangChain fundamentals & ecosystem
- FastAPI basics & async programming
- Development environment setup

Phase 2: Integration (Week 3-4)

- LangGraph for agentic systems
- Advanced LangChain patterns
- FastAPI + LangChain integration

Phase 3: Production (Week 5-6)

- Advanced agentic workflows
 - Deployment & monitoring
 - Performance optimization
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Daily Study Plan (6 Weeks)

WEEK 1: LangChain Foundation

Day 1: Environment Setup & LangChain Basics

Time: 3-4 hours

- ☐ Set up development environment (Python 3.10+, virtual env)
- ☐ Install LangChain, OpenAI/Ollama, Jupyter
- ☐ Read LangChain architecture overview
- ☐ **Hands-on:** Create first simple chain with LLM
- ☐ **Mini-project:** Build a basic Q&A chatbot

Resources:

- [LangChain Documentation](#)

- [LangChain Expression Language \(LCEL\) Guide](#)

Day 2: LangChain Components Deep Dive

Time: 4 hours

- ☐ Study Tools, Agents, and Chains architecture
- ☐ Understand LangChain Expression Language (LCEL)
- ☐ **Hands-on:** Create custom tools and simple agents
- ☐ **Practice:** Build a calculator agent with custom tools

Resources:

- [LangChain Agents Documentation](#)
- [Custom Tools Tutorial](#)

Day 3: LLM Integration & Prompt Engineering

Time: 3-4 hours

- ☐ Set up OpenAI and/or Ollama integration
- ☐ Learn prompt templates and variables
- ☐ **Hands-on:** Create dynamic prompt templates
- ☐ **Practice:** Build a content generation system

Resources:

- [Prompt Engineering Guide](#)
- [LangChain Prompt Templates](#)

Day 4: Output Parsers & Memory Systems

Time: 4 hours

- ☐ Understand different output parsers (JSON, Pydantic)
- ☐ Learn memory types (ConversationBufferMemory, etc.)
- ☐ **Hands-on:** Build a conversational agent with memory
- ☐ **Mini-project:** Create a personal assistant with context

Resources:

- [Output Parsers Documentation](#)
- [Memory Documentation](#)

Day 5: Advanced Chains & Debugging

Time: 4 hours

- ☐ Learn sequential, parallel, and conditional chains
- ☐ Set up logging and debugging workflows
- ☐ **Hands-on:** Build complex multi-step chains
- ☐ **Practice:** Debug common LangChain issues

Resources:

- [Chain Documentation](#)
- [Debugging Guide](#)

Day 6-7: Weekend Project - Document Q&A System

Time: 6-8 hours

- ☐ Build a complete RAG (Retrieval Augmented Generation) system
 - ☐ Implement document loading, chunking, and vectorization
 - ☐ Create conversational interface with memory
 - ☐ Add error handling and logging
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WEEK 2: FastAPI Foundation

Day 8: FastAPI Basics & Project Structure

Time: 3-4 hours

- ☐ Install FastAPI, Uvicorn, and dependencies
- ☐ Learn FastAPI project structure best practices
- ☐ **Hands-on:** Create basic REST API with CRUD operations
- ☐ Understand path parameters, query parameters, request bodies

Resources:

- [FastAPI Tutorial](#)
- [FastAPI Best Practices](#)

Day 9: Async Programming & Database Integration

Time: 4 hours

- ☐ Master async/await patterns in FastAPI

- ☐ Set up SQLAlchemy with async support
- ☐ **Hands-on:** Build async database operations
- ☐ Learn connection pooling and transaction management

Resources:

- [Async SQL \(Relational\) Databases](#)
- [SQLAlchemy Async Tutorial](#)

Day 10: Authentication & Security

Time: 4 hours

- ☐ Implement JWT authentication
- ☐ Learn OAuth2 with FastAPI
- ☐ **Hands-on:** Add authentication to existing API
- ☐ Understand CORS, rate limiting, and security headers

Resources:

- [FastAPI Security](#)
- [OAuth2 with JWT](#)

Day 11: Validation & Error Handling

Time: 3 hours

- ☐ Master Pydantic models for validation
- ☐ Implement custom exception handlers
- ☐ **Hands-on:** Build robust error handling system
- ☐ Learn request/response model patterns

Resources:

- [Pydantic Documentation](#)
- [FastAPI Error Handling](#)

Day 12: Testing & Documentation

Time: 4 hours

- ☐ Write unit tests with pytest and TestClient
- ☐ Learn API documentation with OpenAPI/Swagger
- ☐ **Hands-on:** Test async endpoints and database operations

- ☐ Set up continuous testing workflows

Resources:

- [FastAPI Testing](#)
- [Testing Async Code](#)

Day 13-14: Weekend Project - AI API Service

Time: 6-8 hours

- ☐ Build FastAPI service that integrates your LangChain Q&A system
 - ☐ Add authentication, rate limiting, and proper error handling
 - ☐ Create async endpoints for document upload and querying
 - ☐ Write comprehensive tests and documentation
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WEEK 3: LangGraph & Agentic Systems

Day 15: LangGraph Introduction & Concepts

Time: 4 hours

- ☐ Study LangGraph architecture and state management
- ☐ Understand nodes, edges, and conditional routing
- ☐ **Hands-on:** Build first simple graph workflow
- ☐ Compare with traditional LangChain chains

Resources:

- [LangGraph Documentation](#)
- [LangGraph Tutorials](#)

Day 16: State Management & Persistence

Time: 4 hours

- ☐ Learn LangGraph state schemas and typing
- ☐ Implement persistent state with databases
- ☐ **Hands-on:** Build stateful multi-turn agent
- ☐ Practice state reducers and updates

Resources:

- [State Management Guide](#)

- [Persistence Tutorial](#)

Day 17: Advanced Graph Patterns

Time: 4 hours

- ☐ Learn parallel execution and fan-out/fan-in patterns
- ☐ Implement conditional routing and cycles
- ☐ **Hands-on:** Build complex decision-making agent
- ☐ Study error handling in graph workflows

Resources:

- [Advanced Patterns](#)
- [Conditional Logic Examples](#)

Day 18: Multi-Agent Systems

Time: 4 hours

- ☐ Design multi-agent architectures
- ☐ Implement agent communication protocols
- ☐ **Hands-on:** Build collaborative agent system
- ☐ Learn agent coordination and conflict resolution

Resources:

- [Multi-Agent Systems Tutorial](#)
- Research: "Cooperative Multi-Agent Reinforcement Learning" papers

Day 19: Human-in-the-Loop Systems

Time: 3-4 hours

- ☐ Implement human approval workflows
- ☐ Learn interruption and resumption patterns
- ☐ **Hands-on:** Build human-supervised agent
- ☐ Practice async human interaction handling

Resources:

- [Human-in-the-Loop Tutorial](#)

Day 20-21: Weekend Project - Multi-Agent Research Assistant

Time: 8-10 hours

- ☐ Build system with researcher, analyzer, and writer agents
 - ☐ Implement human approval gates and state persistence
 - ☐ Add real-time monitoring and agent communication logs
 - ☐ Create FastAPI endpoints to control the workflow
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WEEK 4: Advanced Integration

Day 22: LangChain + FastAPI Integration Patterns

Time: 4 hours

- ☐ Learn async integration best practices
- ☐ Implement streaming responses for LLM outputs
- ☐ **Hands-on:** Build streaming chat API
- ☐ Handle long-running LangChain operations

Resources:

- [Streaming Tutorial](#)
- [FastAPI Streaming Response](#)

Day 23: Vector Databases & RAG Systems

Time: 4 hours

- ☐ Integrate Chroma, Pinecone, or Weaviate
- ☐ Implement advanced RAG patterns (HyDE, multi-query)
- ☐ **Hands-on:** Build production RAG API
- ☐ Learn vector similarity search optimization

Resources:

- [Vector Stores Documentation](#)
- [Advanced RAG Techniques](#)

Day 24: Agent Observability & Monitoring

Time: 4 hours

- ☐ Implement LangSmith or custom tracing
- ☐ Set up metrics collection and alerting

- ☐ **Hands-on:** Build agent monitoring dashboard
- ☐ Learn cost tracking and optimization

Resources:

- [LangSmith Documentation](#)
- [Observability Best Practices](#)

Day 25: Performance Optimization

Time: 4 hours

- ☐ Learn LangChain caching strategies
- ☐ Implement connection pooling and batching
- ☐ **Hands-on:** Optimize RAG system performance
- ☐ Profile and benchmark your applications

Resources:

- [Caching Tutorial](#)
- [Performance Tips](#)

Day 26: Advanced FastAPI Features

Time: 4 hours

- ☐ Implement WebSocket support for real-time agents
- ☐ Learn background tasks and job queues
- ☐ **Hands-on:** Add WebSocket chat to your agent system
- ☐ Set up Celery or similar for long-running tasks

Resources:

- [WebSockets](#)
- [Background Tasks](#)

Day 27-28: Weekend Project - Production Agent Platform

Time: 10-12 hours

- ☐ Integrate all learned concepts into a complete platform
- ☐ Build web interface for agent management
- ☐ Implement user management, agent sharing, and monitoring
- ☐ Add comprehensive error handling and logging

WEEK 5: Advanced Topics & Deployment

Day 29: Advanced LangGraph Patterns

Time: 4 hours

- ☐ Implement dynamic graph construction
- ☐ Learn graph optimization and pruning
- ☐ **Hands-on:** Build self-modifying agent system
- ☐ Study hierarchical agent architectures

Resources:

- [Dynamic Graphs](#)
- Research: "Hierarchical Multi-Agent Systems" papers

Day 30: Custom Tools & Extensions

Time: 4 hours

- ☐ Build custom LangChain tools and retrievers
- ☐ Create custom LangGraph nodes and edges
- ☐ **Hands-on:** Develop domain-specific extensions
- ☐ Learn tool versioning and compatibility

Resources:

- [Custom Tools Documentation](#)
- [Extending LangGraph](#)

Day 31: Containerization & Docker

Time: 3 hours

- ☐ Create production Dockerfiles for FastAPI + LangChain
- ☐ Learn multi-stage builds and optimization
- ☐ **Hands-on:** Containerize your agent platform
- ☐ Set up docker-compose for local development

Day 32: Cloud Deployment Strategies

Time: 4 hours

- ☐ Deploy to AWS ECS, Google Cloud Run, or similar

- ☐ Set up load balancing and auto-scaling
- ☐ **Hands-on:** Deploy your platform to the cloud
- ☐ Configure environment variables and secrets management

Day 33: Security & Compliance

Time: 4 hours

- ☐ Implement data privacy and GDPR compliance
- ☐ Learn AI safety and content filtering
- ☐ **Hands-on:** Add security scanning and validation
- ☐ Set up audit logging and compliance reporting

Day 34-35: Weekend Project - Enterprise Agent Platform

Time: 12-15 hours

- ☐ Build multi-tenant agent platform with enterprise features
 - ☐ Implement role-based access control and audit trails
 - ☐ Add analytics dashboard and usage reporting
 - ☐ Create comprehensive deployment documentation
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WEEK 6: Research & Portfolio Development

Day 36: Research Paper Analysis

Time: 4 hours

- ☐ Study recent papers on agentic systems and LLM workflows
- ☐ Analyze ReAct, AutoGPT, and multi-agent research
- ☐ **Practice:** Implement paper concepts in your code
- ☐ Write technical analysis of your implementations

Day 37: Benchmarking & Evaluation

Time: 4 hours

- ☐ Implement agent evaluation frameworks
- ☐ Create custom benchmarks for your use cases
- ☐ **Hands-on:** Evaluate different agent architectures
- ☐ Document performance comparisons

Day 38: Advanced Optimization

Time: 4 hours

- ☐ Learn model fine-tuning for your specific tasks
- ☐ Implement prompt optimization and A/B testing
- ☐ **Hands-on:** Optimize your best-performing agents
- ☐ Study cost-performance trade-offs

Day 39: Portfolio & Documentation

Time: 4 hours

- ☐ Create comprehensive GitHub repositories
- ☐ Write detailed README files and API documentation
- ☐ **Hands-on:** Record demo videos and create examples
- ☐ Set up portfolio website showcasing your work

Day 40: Business Applications

Time: 4 hours

- ☐ Research market opportunities and use cases
- ☐ Create service packages and pricing models
- ☐ **Practice:** Prepare client presentations and proposals
- ☐ Learn compliance requirements for different industries

Day 41-42: Final Integration & Launch

Time: 8-10 hours

- ☐ Complete final integration testing
- ☐ Launch your portfolio and platform
- ☐ Create marketing materials and case studies
- ☐ Plan next steps for business or research development



Project Portfolio (Beginner → Advanced)

Beginner Projects

1. **Personal Q&A Assistant** - RAG system with document upload
2. **Simple Chat API** - FastAPI + LangChain integration
3. **Task Automation Agent** - Basic tool-using agent

Intermediate Projects

4. **Multi-Agent Research System** - Collaborative agents with LangGraph
5. **Real-time Customer Support** - Streaming responses with WebSockets
6. **Content Generation Platform** - Multi-step content creation workflow

Advanced Projects

7. **Enterprise Agent Platform** - Multi-tenant system with monitoring
 8. **Autonomous Business Analyst** - Complex decision-making agents
 9. **Research Paper Implementation** - Novel agentic architecture
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Essential Learning Resources

Core Documentation

- [LangChain Python Docs](#)
- [LangGraph Documentation](#)
- [FastAPI Documentation](#)

Key Research Papers

- "ReAct: Synergizing Reasoning and Acting in Language Models"
- "Toolformer: Language Models Can Teach Themselves to Use Tools"
- "AutoGPT: An Autonomous GPT-4 Experiment"
- "Multi-Agent Systems: A Survey from a Machine Learning Perspective"
- "LangGraph: Multi-Agent Workflows"

GitHub Repositories for Inspiration

- [LangChain Templates](#)
- [LangGraph Examples](#)
- [FastAPI Best Practices](#)
- [Production AI Systems](#)

Courses & Tutorials

- DeepLearning.AI LangChain Course
- FastAPI Course by Testdriven.io
- LangGraph Multi-Agent Systems Course

- Async Python Programming Course
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Career & Business Development

Portfolio Development

- Create 3-5 comprehensive projects showcasing different capabilities
- Write detailed case studies with performance metrics
- Include video demonstrations and live deployed examples
- Contribute to open-source projects in the ecosystem

Freelance Service Packages

1. **AI Chatbot Development** (\$2K-10K)
2. **Document Analysis Systems** (\$5K-20K)
3. **Multi-Agent Automation** (\$10K-50K)
4. **Enterprise AI Platform** (\$25K-100K+)

Research Opportunities

- Contribute to LangChain/LangGraph open source
- Publish papers on novel agent architectures
- Participate in AI research competitions
- Collaborate with academic institutions

Industry Applications

- Customer Service Automation
 - Business Process Intelligence
 - Research & Analysis Platforms
 - Content Creation & Marketing
 - Legal Document Processing
 - Healthcare Data Analysis
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Development Environment Setup

Required Tools

```
bash
```

Core dependencies

pip [install](#) langchain langgraph fastapi uvicorn

pip [install](#) openai anthropic ollama

pip [install](#) sqlalchemy asyncpg alembic

pip [install](#) pytest pytest-asyncio httpx

pip [install](#) pydantic python-jose passlib

pip [install](#) streamlit gradio (for demos)

Vector databases

pip [install](#) chromadb pinecone-client weaviate-client

Monitoring & observability

pip [install](#) langsmith wandb prometheus-client

Deployment

pip [install](#) [docker](#) gunicorn

Recommended IDE Setup

- VS Code with Python, Pylance, and REST Client extensions
- Jupyter Lab for experimentation
- Docker Desktop for containerization
- Postman for API testing



Progress Tracking

Weekly Checkpoints

- ☐ Week 1: LangChain fundamentals mastered
- ☐ Week 2: FastAPI production skills acquired
- ☐ Week 3: LangGraph agent systems operational
- ☐ Week 4: Advanced integration patterns implemented
- ☐ Week 5: Production deployment completed
- ☐ Week 6: Portfolio and business plan ready

Success Metrics

- 9 completed projects in portfolio
- 3 deployed production applications
- Contributions to 2+ open-source projects

- 1 research paper or technical blog post
- Active freelance client pipeline or job interviews

This plan is designed to be modular - you can adjust timelines, skip sections based on prior knowledge, or deep-dive into specific areas of interest. The key is consistent daily practice combined with building real, deployable projects.